Mini Project REPORT

(HARDWARE WORK)

Entitled

"Advance Thermo -Resistive Fire Detection and Alert System"

Submitted in partial fulfillment of the requirement
For the Subject

Electronic Devices & Circuits

(EC108)

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We are really very glad to express our gratitude to our subject co-ordinator, **Prof. Naresh B. Kanirkar**, for giving us his valuable instructions, guidance and continuous support which extremely helped us in completing our Fire Alarm Circuit project successfully.

We would also like to thank all the staff members and teaching assistants who gave their valuable suggestions for selecting and getting the specific rated components used in our project.

This mini project has helped us a lot in gaining practical knowledge and a better understanding of Electronic Devices & Circuits, which will surely be useful for us in the future.

Nishant Tiwari

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Abstract

Fire detection and safety is very important in homes, laboratories, industries, hostels and public places. A fire can spread very quickly, so an early warning system is necessary to alert people and prevent major loss.

In this project, we have designed a simple **Fire Alarm Circuit** using basic electronic components. The circuit consists of an NTC Thermistor (heat sensor), BC547 NPN transistor (as a switch), 10KΩ potentiometer (for sensitivity adjustment), piezo buzzer, optional LED and a 9V battery. The entire circuit is built on a General Purpose PCB (perfboard) with permanent soldering.

Monitoring for fire can be done in two ways: one is manually checking for smoke or heat, and the other is to use an automatic electronic fire alarm. The easier and faster way is to use a Fire Alarm Circuit. It comes in different shapes and sizes in real life, but the simplest version for student projects is the temperature-based alarm using a thermistor.

There are many expensive fire alarm systems available in the market which use smoke sensors and microprocessors. Here, the point of selecting the above-mentioned Fire Alarm Circuit was due to the fact that it is very easy to construct, uses very few and low-cost components, and gives a clear understanding of how electronic devices and circuits work in real applications.

This project helped me learn the principle of NTC thermistor, transistor switching action, and how a small change in resistance can control a buzzer and LED. The circuit basically works on a single 9V battery. The output is given by a loud buzzer sound and an optional glowing LED, indicating the presence of fire/heat.

The circuit is built on a General Purpose PCB and tested successfully. It is a good prototype for basic fire detection and gives instant audio-visual alert when temperature rises.

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CHAPTER-1

INTRODUCTION

Fire detection is very important for safety in homes, laboratories, industries and public places. A fire alarm circuit detects sudden rise in temperature (heat) and gives an audio-visual alert so that people can evacuate or take action immediately.

In this project, we have designed a simple Fire Alarm Circuit using an NTC Thermistor as the heat sensor, BC547 transistor as a switch, a 10K potentiometer for sensitivity adjustment, a piezo buzzer and a 9V battery. The circuit works on basic analog electronics principles without any microcontroller or complex IC.

There are many expensive fire alarm systems available in the market which use smoke sensors and microprocessors. But the most commonly used basic fire alarm for student projects is the temperature-based one using a thermistor. The point of selecting this circuit was due to the fact that it is easy to construct, uses very few components and gives a clear understanding of how electronic devices and circuits work in real life.

This project helps us learn the principle of thermistors, transistor switching action, and how a small change in resistance can control a buzzer. The output is given by a loud buzzer and an optional LED indicator. The circuit basically works on a single 9V battery and is built on a General Purpose PCB (perfboard) for permanent soldering.

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COMPONENTS AND THEIR DESCRIPTION

Sr. no.	Component name	Quantity
1	BC547 NPN Transistor	1
2	NTC Thermistor (10KΩ)	1
3	10KΩ Potentiometer	1
4	Piezo Buzzer (5-12V)	1
5	LED (Red)	1
6	9V Battery + Clip	1
7	Resistors (220Ω)	1
8	General Circuit Board (PCB)	1
9	Jumper wires & Bread Board	As required
10	Soldering Iron, Wire, Multimeter	For assembly

Description of main components:

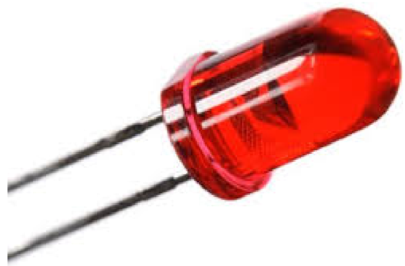
1. **BC547 NPN Transistor** It is a general-purpose NPN bipolar junction transistor used as a switch in this circuit. When sufficient voltage is applied to its base, it turns ON and allows current to flow through the collector-emitter path, activating the buzzer.

2. **NTC Thermistor (10KΩ)** Negative Temperature Coefficient thermistor. Its resistance is high at room temperature and decreases sharply when heated. This change in resistance is used to detect fire/heat.

3. **10KΩ Potentiometer** A variable resistor used to adjust the sensitivity of the circuit so that the alarm does not trigger at normal room temperature but only when heat is detected.

4. **Piezo Buzzer** An active buzzer that produces a loud sound when voltage is applied. It gives the audio alarm.

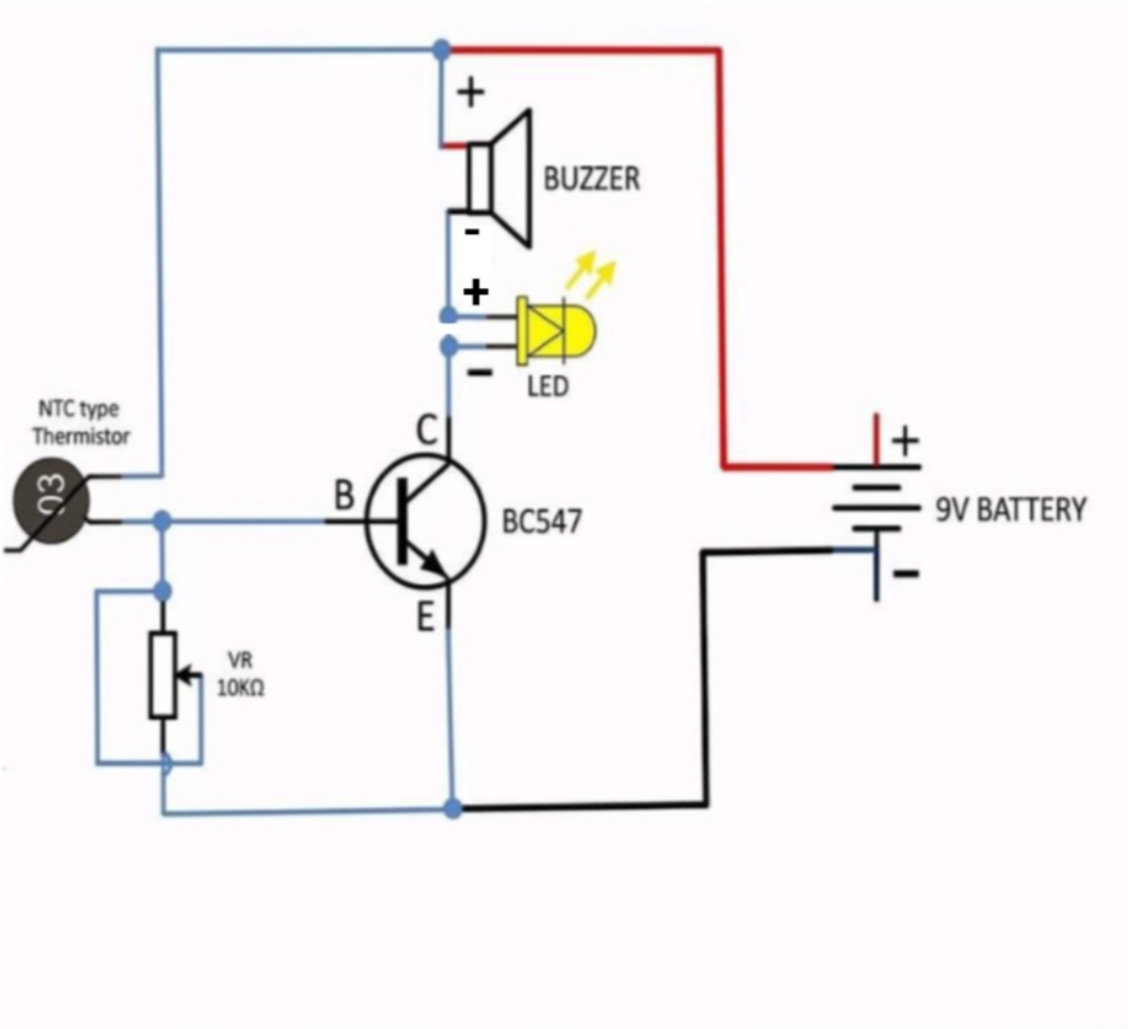
5. **LED** Optional visual indicator that glows along with the buzzer when the alarm is triggered.



6. **9V Battery** Single power source for the entire circuit.

CHAPTER 3

Circuit Diagram



CHAPTER 4

Working

The principle behind the working of this Fire Alarm is the change in resistance of NTC Thermistor with temperature.

When the thermistor is at room temperature, its resistance is high. Therefore, very little current flows to the base of BC547 transistor and it remains OFF. The buzzer stays silent.

When heat (fire) comes near the thermistor, its resistance drops sharply. This increases the voltage at the base of the transistor. The transistor turns ON and acts as a closed switch. Current flows through the collector to the buzzer, making it sound loudly. The optional LED also glows, giving visual indication.

The 10K potentiometer is used to set the exact trigger point so that the circuit does not false-trigger at normal temperature but responds only to heat.

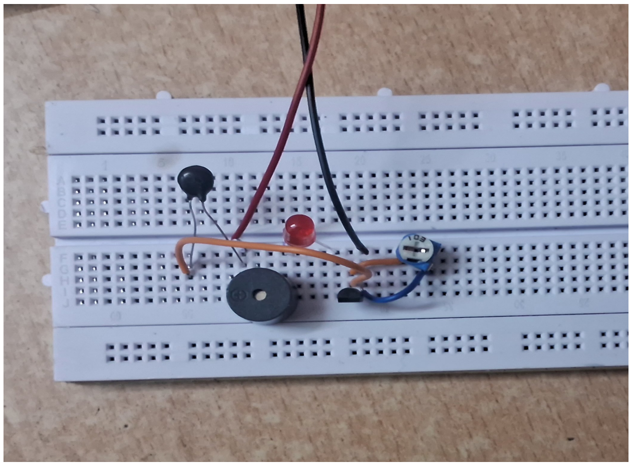
Useful Steps for Assembly on General PCB:

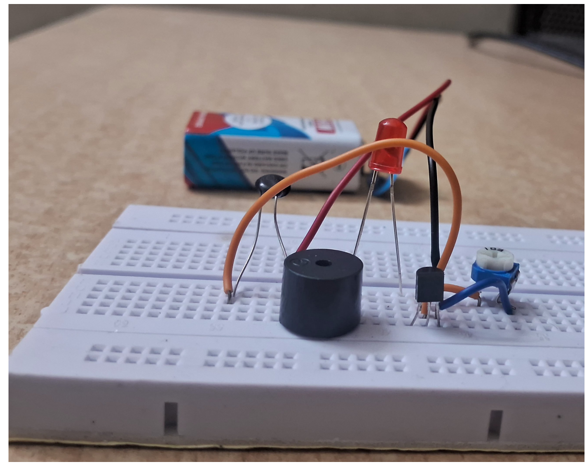
1. Solder the BC547 transistor on the PCB.
2. Solder the NTC thermistor, potentiometer, buzzer and LED.
3. Connect all resistors and wires as per the circuit.
4. Solder the 9V battery clip.
5. Test the circuit by applying heat (lighter/matchstick near thermistor – do not touch).

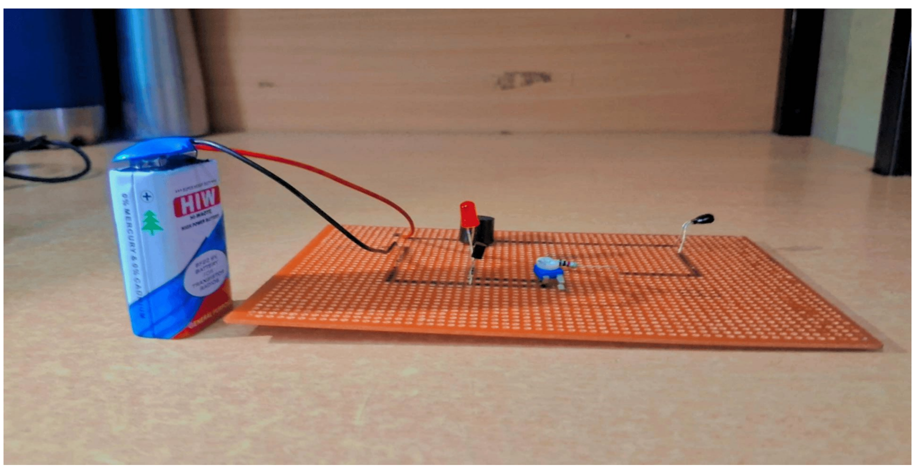
The rate at which the buzzer sounds shows the presence of heat/fire. This circuit can work as a basic prototype for fire detection and gives a clear understanding of transistor switching and sensor applications.

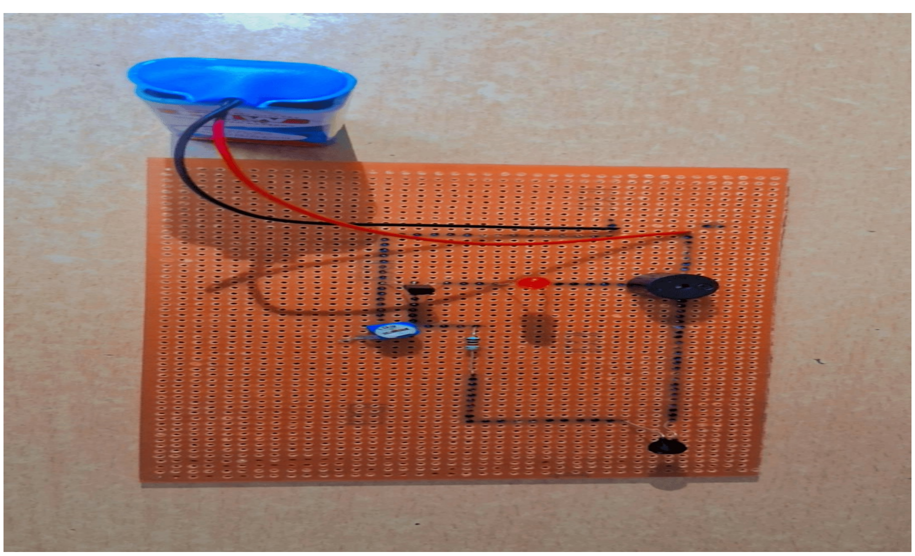
CHAPTER 5

Photo Gallery









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ADVANTAGES

- Very simple and low-cost circuit (total cost under ₹250).
- Easy to understand and build on General PCB – perfect for learning Electronic Devices & Circuits.
- No microcontroller or programming required.
- Uses only basic components (thermistor + transistor).
- Sensitivity can be easily adjusted with potentiometer.
- Gives both audio (buzzer) and visual (LED) alarm.
- Works instantly on 9V battery and is portable.
- Good prototype for real-life fire detection systems

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CONCLUSION

The simple Fire Alarm Circuit using NTC thermistor and BC547 transistor was successfully designed, built and tested on a General Purpose PCB. The project works perfectly – the buzzer sounds loudly when heat is detected and stops when heat is removed. The potentiometer allows precise sensitivity adjustment.

This mini project helped me gain practical knowledge of thermistors, transistor switching action, biasing, and soldering on perfboard. It also gave a clear idea of how basic electronic components can be used to solve real-world safety problems.

In future, this circuit can be upgraded by adding a smoke sensor (MQ-2) or Arduino for digital display of temperature.

Overall, the project was a great learning experience in the subject “Electronic Devices & Circuits”.